PROBLEM 05

Plane and Glider

Original statement · Problems.pdf, p. 7

Aerospace Towed Flight

An airplane A is flying horizontally with a constant speed v_A, and is towing a glider B. The glider is gaining altitude. The tow cable has a length r, and the angle θ formed between the tow cable and the horizontal is increasing at a constant rate of $\dot{\theta}$. We want to determine the magnitude of the velocity, v_B, and the acceleration, a_B, of the glider as a function of θ and its derivatives.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Tow plane A flies horizontally at constant speed v_A. Glider B is towed behind on a cable of constant length r. The tow angle $\theta(t)$ above the horizontal increases at rate $\dot{\theta}(t)$.
- **Hypotheses:** Cable length $r = \text{const}$, plane speed $v_A = \text{const}$ (so $\dot{v}_A = 0, \mathbf{a}_A = \mathbf{0}$).
- **Parameters & Units:** Plane speed v_A $[\text{m/s}]$, cable length r $[\text{m}]$, angle θ $[\text{rad}]$, angular rate $\dot{\theta}$ $[\text{rad/s}]$, angular acceleration $\ddot{\theta}$ $[\text{rad/s}^2]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$, with $\mathbf{i}_0$ pointing horizontally in plane A's direction of flight, and $\mathbf{j}_0$ pointing vertically upward.
- **Position of Plane A:**

$$\mathbf{r}_A(t) = (x_{A0} + v_A t)\mathbf{i}_0$$

- **Relative Position of Glider B with Respect to A:**

Since the glider is behind A and gaining altitude while θ increases, the vector from A to B points at angle θ above the negative horizontal:

$$\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

The total position of glider B in $\mathcal{S}_0$ is:

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = (x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + (r\sin\theta)\mathbf{j}_0$$

- **Erratum:** The first version said that the vector from A to B is at angle θ below the negative horizontal, which contradicts $\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$ (positive vertical component). The glider is behind and above A, at angle θ above the negative horizontal; this is the only reading compatible with the glider gaining altitude as θ increases, and the formulas for $\mathbf{v}_{B/0}$ and $\mathbf{a}_{B/0}$ are unchanged. Numeric check: the altitude $r\sin\theta$ has derivative $r\dot{\theta}\cos\theta > 0$ for $0 < \theta < \pi/2$ and $\dot{\theta} > 0$.

PHASE 3

Step-by-Step Mathematical Deduction

Velocity of Glider B

Pedagogical Justification: The absolute velocity is obtained by differentiating position $\mathbf{r}_B$ in the inertial frame $\mathcal{S}_0$:

$$\mathbf{v}_{B/0} = \frac{d\mathbf{r}_B}{dt} = \frac{d}{dt}(x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + \frac{d}{dt}(r\sin\theta)\mathbf{j}_0$$

- **Explicit Chain Rule:**

$$\frac{d}{dt}(-r\cos\theta) = -r(-\sin\theta)\dot{\theta} = r\dot{\theta}\sin\theta$$

$$\frac{d}{dt}(r\sin\theta) = r\cos\theta\,\dot{\theta}$$

Therefore:

$$\mathbf{v}_{B/0} = (v_A + r\dot{\theta}\sin\theta)\mathbf{i}_0 + (r\dot{\theta}\cos\theta)\mathbf{j}_0$$

- **Magnitude of Velocity $v_{B/0}$:**

$$v_{B/0} = \|\mathbf{v}_{B/0}\| = \sqrt{(v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

Acceleration of Glider B

> **Pedagogical Justification:** Acceleration is the time derivative of velocity $\mathbf{a}_{B/0} = \frac{d\mathbf{v}_{B/0}}{dt}$.

- **Explicit Product & Chain Rules for Each Component:**
- x-component differentiation:

$$\frac{d}{dt}(v_A + r\dot{\theta}\sin\theta) = 0 + r\frac{d}{dt}(\dot{\theta}\sin\theta) = r(\ddot{\theta}\sin\theta + \dot{\theta}(\cos\theta\dot{\theta})) = r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta$$

- y-component differentiation:

$$\frac{d}{dt}(r\dot{\theta}\cos\theta) = r\frac{d}{dt}(\dot{\theta}\cos\theta) = r(\ddot{\theta}\cos\theta + \dot{\theta}(-\sin\theta\dot{\theta})) = r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta$$

- **Vector Form:**

$$\mathbf{a}_{B/0} = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)\mathbf{i}_0 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)\mathbf{j}_0$$

- **Magnitude of Acceleration $a_{B/0}$:**

$$a_{B/0}^2 = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)^2 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)^2$$

Expanding both squared terms:

$$= r^2\ddot{\theta}^2\sin^2\theta + 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\cos^2\theta$$

$$+r^2\ddot{\theta}^2\cos^2\theta - 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\sin^2\theta$$

The cross terms cancel out completely:

$$a_{B/0}^2 = r^2\ddot{\theta}^2(\sin^2\theta + \cos^2\theta) + r^2\dot{\theta}^4(\cos^2\theta + \sin^2\theta) = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$$

Taking the square root:

$$a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

PHASE 4

Physical Interpretation & Dimensional Verification

- **Plane Motion Decoupling:** Notice that the acceleration magnitude $a_{B/0}$ is completely independent of the tow plane's cruise velocity v_A because $\mathbf{a}_A = \mathbf{0}$ (uniform rectilinear translation). The acceleration felt by the glider is purely relative circular motion around A!
- **Limiting Case $\ddot{\theta} = 0$:** If the cable turns at a constant angular rate $\dot{\theta} = \Omega$, the acceleration reduces to pure centripetal acceleration: $a_{B/0} = \sqrt{r^2\Omega^4} = r\Omega^2$.
- **Dimensional check:** $[r\ddot{\theta}] = [r\dot{\theta}^2] = \mathrm{m}\cdot\mathrm{s}^{-2}$ and $[v_A + r\dot{\theta}\sin\theta] = \mathrm{m/s}$, so $a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$ is in $\mathrm{m/s^2}$ and $v_{B/0}$ in m/s.
- **Limit $\dot{\theta} = 0$:** $\mathbf{v}_{B/0} = v_A\,\mathbf{i}_0$ and $\mathbf{a}_{B/0} = \mathbf{0}$, the glider moves rigidly with the tow plane.
- **Symbolic check (sympy):** differentiating $\mathbf{r}_B$ twice for an arbitrary function $\theta(t)$ gives $\|\mathbf{v}_{B/0}\|^2 = (v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2$ and $\|\mathbf{a}_{B/0}\|^2 = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$ with zero residual.

Source: Problems.pdf